

Voluntary Report – Voluntary - Public Distribution

Date: July 17, 2026

Report Number: E42026-0057

Report Name: Biofuel Mandates in the EU by Member State - 2026

Country: European Union

Post: Berlin

Report Category: Biofuels

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Report Highlights:

This report provides an overview of the biofuel use mandates in EU-27 member states, including national implementation of the revised RED II in some Member States. It supplements the EU Biofuels Annual Report for 2026.

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Abbreviations and Definitions Used in this Report

% Cal =	percent energy content
% Vol =	percent volume
% Biodiesel =	minimum percentage of biodiesel in total diesel use
% Bioethanol =	minimum percentage of bioethanol in total gasoline use
% Overall =	minimum percentage of biofuels in total fuel use
% GHG Emission Reduction =	Percentage of GHG reductions of total fuel use (fossil and renewable) compared to the hypothetical GHG emissions that would have occurred with the exclusive use of fossil fuels

All of the above refer to fuel use in the transport sector.

Advanced biofuel =	Non-food and feed-based biofuels that are listed in Annex IX of the Renewable Energy Directive.
Biodiesel =	Fatty acid methyl ester produced from agricultural or waste feedstock (vegetable oils, animal fat, recycled cooking oils) used as transport fuel to substitute for petroleum diesel
Bioethanol =	Ethanol produced from agricultural feedstock used as transport fuel
Cat. 1, 2, and 3 =	Risk categories for animal by-products as defined in EU Regulation (EC) 1069/2009 , with cat. 1 having the highest and cat. 3 the lowest risk.
Double counting =	Certain biofuels are counted twice against mandates. This was introduced to support the use of certain biofuels and/or feedstocks. As a result of double counting, less physical volumes of a certain biofuel are needed to fill a mandate, which makes the respective biofuel more attractive than a comparable single-counting biofuel. Definition and eligible feedstocks vary by member state (MS).
EC =	European Community or European Commission - depending on the context
ETBE =	Ethyl tert-butyl ether, an oxygenate gasoline additive containing 47% vol ethanol
EU =	European Union
FQD =	EU Fuel Quality Directive 98/70/EC amended by directives 2009/30/EC and (EU) 2015/1513
GHG =	Greenhouse gas
GJ =	Gigajoule = 1,000,000,000 Joule or 1 million KJ
ILUC =	Indirect land use change
Ktoe =	1000 MT of oil equivalent = 41,868 GJ = 11.63 GWh
MJ =	Megajoule
MS =	Member State(s) of the EU
MT =	Metric ton
MWh =	Mega Watt hours = 1,000 Kilo Watt hours (KWh)
N/A =	Not applicable
PFAD =	Palm fatty acid distillate
POME =	Palm oil mill effluent
RED =	EU Renewable Energy Directive 2009/28/EC
RED II =	EU Renewable Energy Directive 2018/2001/EC

RES =	Renewable energy sources
RES-T =	Renewable energy share in transport
RFNBO =	Renewable fuels of non-biological origin
SAF =	Sustainable aviation fuel
SBE =	Spent bleached earth
Tall oil =	A by-product of the wood manufacturing industry; qualifies as advanced biofuels feedstock
Tall-oil pitch =	The residue from the distillation of tall oil; qualifies as advanced biofuels feedstock
TME =	Tallow Methyl Ester, biodiesel made from animal fat
Toe =	Tons of oil equivalent = 41,868 MJ = 11.63 MWh
UCO =	Used cooking oil/ recycled vegetable oil
UCOME =	UCO based methyl ester biodiesel
UER =	Upstream emission reduction

Introduction

The European Union (EU) adopted the second iteration of the Renewable Energy Directive (REDII, [Directive 2018/2001](#)) covering the period 2021-2030 in 2018. Most of the provisions of this directive entered into force on January 1, 2021. The directive was amended in October 2023 by [Directive 2023/2413](#) to align REDII with the EU's Green Deal goal of reducing greenhouse gas emissions to 55 percent by 2030 (compared to 1990) and achieving carbon neutrality by 2050. The revised REDII entered into force on November 20, 2023, with an 18-month period to transpose most of the directive's provisions into Member States (MS) national law. So far, most Member States are still in the transposition process.

Many Member States have adopted minimum biofuel use mandates in order to achieve the goals of RED and RED II. This report provides an overview of current and future mandates from the various Member States. For information on mandates referring to years prior to 2022 please check our report from 2022.¹ Note that Cyprus, Luxemburg, and Malta are not included in this report. The tables represent the status quo as of June 2026. Changes, especially pertaining to the revised RED II, that have not officially been adopted into national law by Member States yet are mentioned in the text.

EU Renewable Energy Targets

The revised RED II sets out a target for overall renewable energy share of at least 42.5 percent binding at the EU level by 2030. For transport, Member States can choose between a target of reducing greenhouse gas (GHG) intensity by 14.5 percent up to 2030 or ensuring a share of at least 29 percent of renewables in final energy consumption by 2030. The Directive also sets out a binding target on non-crop-based biofuels of 1 percent in 2025 and 5.5 percent in 2030, of which a share of at least 1 percentage point is from renewable fuels of non-biological origin in 2030. The EU capped crop-based biofuels at the level consumed in each Member State in 2020, with an additional 1 percent allowed over present consumption up to an overall maximum cap of 7 percent. Member States can also set a lower limit for conventional biofuels than prescribed in RED II. For advanced biofuels, RED II introduces two different sets of targets for feedstock listed in Part A of Annex IX and feedstock listed in Part B. Feedstock listed in Part A must be supplied at a minimum 1 percent in 2025 and 5.5 percent in 2030, of which a share of at least 1 percent is from renewable fuels of non-biological origin (RFNBO) in 2030. Biofuels produced from feedstock listed in Part B will be capped at 1.7 percent in 2030 except in Cyprus and Malta. Advanced biofuels can be double counted.

¹ [Biofuel Mandates in the EU by Member State - 2022 Berlin European Union E42022-0044](#)

**Advanced Biofuel Sources,
Part A and Part B of Annex IX**

Part A	Part B
• Algae if cultivated on land in ponds or photobioreactors • Biomass fraction of mixed municipal waste • Biowaste from private households subject to separate collection • Biomass fraction of industrial waste not fit for use in the food or feed chain • Straw • Animal manure and sewage sludge • Palm oil mill effluent and empty palm fruit bunches • Crude glycerin • Bagasse • Grape marcs and wine lees • Nut shells • Husks • Cobs cleaned of kernels of corn • Biomass fraction of wastes and residues from forestry and forest-based industries • Other non-food cellulosic material • Other ligno-cellulosic material except saw logs and veneer logs • Fusel oils from alcoholic distillation • Raw methanol from kraft pulping stemming from the production of wood pulp • Intermediate crops, such as catch crops and cover crops that are grown in areas where due to a short vegetation period the production of food and feed crops is limited to one harvest and provided their use does not trigger demand for additional land, and provided the soil organic matter content is maintained, where used for the production of biofuel for the aviation sector • Crops grown on severely degraded land, except food and feed crops, where used for the production of biofuel for the aviation sector • Cyanobacteria	• Used cooking oil (UCO) • Some categories of animal fats • Damaged crops that are not fit for use in the food or feed chain, excluding substances that have been intentionally modified or contaminated in order to meet this definition • Municipal wastewater and derivatives other than sewage sludge • Crops grown on severely degraded land excluding food and feed crops and feedstocks listed in Part A of this Annex, where not used for the production of biofuel for the aviation sector • Intermediate crops, such as catch crops and cover crops, and excluding feedstocks listed in Part A of this Annex, that are grown in areas where due to a short vegetation period the production of food and feed crops is limited to one harvest and provided their use does not trigger demand for additional land and provided the soil organic matter content is maintained, where not used for the production of biofuel for the aviation sector

A non-exhaustive list of waste and residues covered by Annex IX of RED II can be found in Annex IV of [Implementing Regulation 2022/996](#) on rules to verify sustainability and greenhouse gas emissions saving criteria and low indirect land-use change (ILUC) risk criteria. The substances listed in that annex

shall be considered as falling under a category of raw material set out in Annex IX without being explicitly mentioned.

Advanced Biofuel Sources, Part A and Part B of Annex IX, Mandates and Cap		
	Part A Mandates (% cal)	Part B Cap (% cal)
2022	0.2	-
2025	1	-
2030	3.5	1.7

In the revised RED II, advanced biofuels from feedstock listed in Annex IX and renewable fuels of non-biological origin can still be double counted towards the targets.

EU-wide Greenhouse Gas (GHG) Emission Reductions

The [Renewable Energy Directive \(RED\)](#) stipulated that biofuels can only be counted against EU and/or member state targets if they fulfill the following minimum greenhouse gas (GHG) reduction requirements:

	RED Minimum % GHG Emission Reductions of Each Biofuel Compared to the Respective Fossil Fuel
2009-2017	35%
2018 and onwards	50% for biofuels produced in operations that started production on or before Oct 5, 2015. 60% for biofuels produced in operations that started production after Oct 5, 2015.

Source: Art. 7 b of [EU Directive 98/70/EC](#) as revised by [Directive \(EU\) 2015/1513](#)

The **Fuel Quality Directive (FQD)** ([Directive 2009/30/EC](#)) complemented RED and mirrored some of RED's content such as sustainability criteria. The FQD limits ethanol blends to 10 percent or less when ethanol is used as an oxygenate, and places limits on palm oil and soy oil content of biodiesel. The Commission did not increase the GHG reduction target in the FQD for the time beyond 2020. Instead, the Commission addressed the issue of decarbonization of transport fuels after 2020 in RED II.

In October 2023, the EU adopted [Regulation 2023/2405](#) on Sustainable Aviation Fuels (SAF). The Regulation requires aviation fuel suppliers to ensure all aviation fuel made available to aircraft operators at each EU airport contains a progressively increasing minimum share of SAF, including a minimum share of synthetic aviation fuels (renewable fuels of non-biological origin), in accordance with the values and dates of application set below:

Targets in SAF Regulation (volume based)

Date of Application	Minimum Share of SAF	Minimum Share of Synthetic Fuels
January 1, 2025	2%	N/A
January 1, 2030	6%	1.2%
January 1, 2035	20%	5%
January 1, 2040	34%	10%
January 1, 2045	42%	15%
January 1, 2050	70%	35%

The EC defines SAF as aviation fuels that are either synthetic aviation fuels, recycled carbon aviation fuel, advanced biofuels as listed from feedstock listed in part A and B of Annex IX or biofuels produced from feedstock other than food and feed crops which comply with sustainability and GHG emissions criteria from RED II. Therefore, crop-based biofuels cannot count towards the targets set for SAF.

RED II introduced new GHG emission criteria that biofuels used in transport must comply with to be counted towards the overall 14 percent target. The European Commission is allowed to revise and update the default values of GHG emissions when technological developments make it necessary. Economic operators have the option to either use default GHG intensity values provided in RED II or to calculate actual values for their pathway.

Greenhouse Gas Saving Thresholds in RED II			
Plant Operation Start Date	Transport Biofuels	Transport Renewable Fuels of Non-biological Origin	Electricity, Heating, and Cooling
Before October 2015	50%	-	-
After October 2015	60%	-	-
After January 2021	65%	70%	70%
After January 2026	65%	70%	80%

Mandates by Member State:

To provide context, current, expired, and future mandates are listed below, by member state in alphabetical order. Mandates based on energy content are expressed in % cal, volume-based mandates in % vol, and GHG-saving mandates in % GHG emission reduction (compared to the hypothetical GHG emissions that would have occurred with the exclusive use of fossil fuels). **For easy reference, mandates applicable in 2026 are bold.**

Cap of Crop-Based Biofuels by Member State

Crop-based Cap (% cal)

Country	2026	2027	2028	2029	2030
Austria	7				
Belgium	7				
Bulgaria	7	7	7	7	7
Croatia					
Czechia	5.6				
Denmark	4,5				
Estonia	0.5		-		
Finland	2.6; High ILUC: 0				
France	7	7	7	7	7
Germany	4.9	4.9	4.6	4.7	4.9
Greece	7	7	7	7	7
Hungary	7	7	7	7	7
Ireland	3	3	3	3	3
Italy					
Latvia	-				
Lithuania	No more than 1.0% higher than the total share in 2020				
The Netherlands	1.2				
Poland	6.1				
Portugal	3.1				
Romania	7	7	7	7	7
Slovakia	6.5				
Slovenia	7				
Spain	2.6				
Sweden	7	7	7	7	7

Austria

	Overall Percentage (energy content, % cal)	Biodiesel (% cal)	Bioethanol (% cal)	Advanced Biofuels (% cal)	GHG Emission Reduction (%) ¹⁾	Cap on Crop-based Biofuels (% cal)	Multiple Counting
2024	None	6.3	3.4	0.2	7	7 ²⁾	No
2025				1	7.5		
2026				1	8		
2027				1	9		

2028				1	10		
2029				1	11		
2030				3.5	13		

Source: FAS Vienna based on [Austrian Fuels Order 2012](#), (with its 2014, 2017, 2018, 2020, 2022, and 2024 amendments)

- 1) To reach the GHG reduction target, the following may be taken into account:
Electric power from renewable energy sources used for electrically powered motor vehicles may also be taken into account (multiple counting x4 for renewable electricity in road transport).
- 2) Palm oil-based biofuels are excluded since July 1, 2021.

Penalties

Failure to comply with these measures could result in the following penalties:

Mandate	Penalty
Energy	€ 43 per GJ under supplied
GHG reduction 2024 and onwards	€ 600 per MT CO ₂ eq of unmet GHG reduction target

Tax incentive

In addition to penalties for non-compliance, there is a tax incentive for biofuels. For gasoline with a minimum content of biogenic substances of 46 per 1,000 liters, the reduced mineral oil tax is € 482 per 1,000 liters (regular tax rate = € 515). For diesel with a minimum content of biogenic substances of 66 liters per 1,000 liters, the reduced mineral oil tax is € 397 per 1,000 liters (regular tax rate = € 425). Pure biofuels in transportation are fully exempt from the mineral oil tax.

Belgium

	Overall Percentage (% cal)	Biodiesel (% cal)	Bioethanol (% cal)	Double Counting
2022	10.2	6.5	6.5	Max 0.95 %
Since January 1, 2023	10.5	5.7	5.7	Max 0.95%

Source: FAS USEU based on [Law of July 7, 2013](#); [Law of July 21, 2017](#); *Law of May 4, 2018*; *Law of December 27, 2021*

Since the increase of the bioethanol mandate at the beginning of 2017, the majority of Belgian gasoline is E10, with the exception of a remnant market for bioethanol-free gasoline for older cars and small engines like lawnmowers.

Penalties

Failure to comply with these measures could result in the following penalties:

Mandate	Penalty
Energy	€1400 per 34 GJ undersupplied.

Source: [ePure](#)

Bulgaria

Biodiesel (% vol)		Bioethanol (% vol)		Advanced Biofuels (min. % energy content)	Cap on crop-based biofuels (% vol)	Multiple Counting
April 1, 2019 – August 31, 2024	6	March 1, 2019 - February 28, 2025	9	April 2019 - December 2021 0.05%	7	No
	Incl.1% advanced biodiesel			2022 - 2024 0.2%		
From September 1, 2024 onwards	6 Incl. 2% advanced biodiesel	From March 1, 2025 onwards	9 ¹⁾ Incl. 1% advanced bioethanol	2025 - 2029 1.0%		
				As of 2030 3.5%		

Source: FAS Sofia

- 1) From March 1, 2025, the mandate for bioethanol was raised to 9 percent, with at least 1 percent of the volume of the bioethanol composed of advanced bioethanol (Art.47/7)

Note: In the period between December 31, 2023, to December 31, 2030, the share of biofuels produced from high ILUC-risk feedstocks is reduced to zero. Calculation of energy consumption of biofuels in transport produced from feedstocks of type B, is limited to 1.7 percent of the energy content of fuels in transportation.

In 2018, Bulgaria's Parliament amended the nation's *Renewable Energy Law* to transpose [Directive 2015/1513/EU](#) into national law. The legislation introduced sustainability criteria (traceability) for advanced biofuels. Adopted changes are in Article 47 of the *Bulgarian Renewable Energy Law* that entered into force on April 1, 2019. The institution supervising the quality of fuel, biofuel mandates, and advanced biofuels on the market is the Agency for Metrological and Technical Surveillance under the Ministry of Economy. In 2025, Bulgaria fully transposed RED II, a process which was delayed due to frequent political changes. In early 2026, the Parliament reviewed the transposition of revised RED II, however, the bill was not voted due to early elections in April. Currently, the newly elected Government and Parliament are ready to finalize the legislation harmonization, and it is expected that the national [Renewable Energy Law](#) (note: current version as of November 2025) will be fully compliant before the end of 2026.

Per the current legislation, the maximum blend with conventional fuel cannot exceed 7 percent for biodiesel, and 10 percent for bioethanol, to be in line with the EU's FQD. The crop-based biofuels cap is at 7 percent, excluding biofuels made from raw materials which are compliant with sustainability criteria and are grown on degraded or fallow land. Biofuels should make up a 9 percent share in petrol and a 6 percent share in diesel, in volume.

The minimum targets for advanced biofuels in transport (Annex IX-A in RED II) are described in Art. 13 and are set at 0.2 percent in 2022 (and in 2023 and 2024), 1 percent in 2025 and 3.5 percent in 2030. For the period 2019 – 2021, the target was at 0.05 percent, which was fulfilled mainly by advanced biodiesel due to lack of local production of advanced bioethanol. These targets must be achieved by mandates for the market players described in Art. 47 of the current [Renewable Energy Law](#) (see table). From September 2024, 2 percent of the volume of biodiesel should be advanced biodiesel and as of March 2025, 1 percent of the volume of bioethanol should be advanced bioethanol.

Croatia

National Targets for Renewable Energy in Transport

Year	National Goal (in %)	Realized (in %)	Public Sector Share (in %)	Share of the Obligated (Total, in %)	Electricity from RES (in %)	Renewable Fuels of Non-biological Origin (RFNBO) (in %)	Advanced Biofuels (in %)	Biofuels from UCO (in %)	1st Gen Biofuels (in %)
2021	7.1	7.1	1.2	5.9	0	0	0	3.4	2.5
2022	8.1	2.4	1.5	6.6	0.2	0	0.4	3.5	2.6
2023	8.8		1.8	7.1	0.2	0	0.8	3.5	2.6
2024	9.5		1.9	7.6	0.4	0	1.1	3.4	2.7
2025	10.5		2.0	8.5	0.8	0.1	1.5	3.4	2.7
2026	12.0		2.3	9.7	1.3	0.4	1.9	3.4	2.7
2027	16.1		2.6	13.5	2.1	0.8	4.6	3.4	2.7
2028	18.5		2.9	15.6	3.1	1.3	5.1	3.4	2.7
2029	21.2		3.3	17.9	4.3	1.9	5.6	3.4	2.7
2030	24.6		3.8	20.8	5.8	2.7	6.2	3.4	2.7

Source: *The Integrated National Energy and Climate Plan (NECP) for the Republic of Croatia, Final Updated (2021-2030)*.

The *Law on the Act on Biofuels for Transport* entered into force in 2009 and was last amended on May 22, 2021. This Croatian law regulates biofuels and renewable energy sources in transportation. It is designed to transpose EU directives into Croatian national legislation. According to the *Integrated National Energy and Climate Plan for the Republic of Croatia, Final Updated (2021-2030)* (NECP), Croatia aims to have a share of renewable energy sources (RES) in gross final energy consumption at 42.5 percent and the share of RES in final energy consumption in transport at 24.6 percent by 2030.

Penalties are defined by the Government Decree on Penalties for the Environment for not Placing Biofuels on the Market and for not Reducing Greenhouse Gas Emissions.

Penalties

Failure to comply with these measures could result in the following penalties:

Mandate	Penalty
Energy % from Biofuels 0%-3%	Kn 0.30 per MJ (0.04€) under supplied
Energy % from Biofuels 3.01%-5%	Kn 0.15 per MJ (0.02€) under supplied
Energy % from Biofuels >5%	Kn 0.01 per MJ (0.0013€) under supplied
GHG reduction 0%-2%	Kn 3.5 per kg CO ₂ (0.46€) under allocated
GHG reduction 2.01%-6%	Kn 0.01 per kg CO ₂ (0.013€) under allocated

Czechia

Czechia has transposed the revised RED II into national legislation. The process of incorporating Directive (EU) 2023/2413 of the European Parliament and of the Council (revised RED II) into national legislation was completed by the adoption of *Act 42/2025 Coll.* of January 22, 2025, and *Act 87/2025 Coll.* of March 4, 2025. *Act 42/2025* amends *Act No. 201/2012 Coll.*, on air protection, as amended (hereinafter the amended Act on air protection). *Act 87/2025 Coll.* amends *Act No. 458/2000 Coll.* on the conditions of doing business and on the performance of state administration in the energy sector and on the amendment of certain acts (Energy Act), as amended, and other related acts, including the amendment to the Act on Supported Sources (hereinafter the amendment to the Act on Supported Energy Sources). Both laws take into account the extension of the scope of the Act to all fuel suppliers. The Ministry of the Environment is responsible for the Air Protection Act. Its amendment partially transposes the revised RED II Directive and will be completed by issuing related accompanying legal regulations.

National obligations and mandates of fuel suppliers and measures for the use of renewable energy sources in the transport sector in Czechia until 2030 and onwards are below.

Year	Advanced liquid and gaseous renewable fuels from feedstocks IX.A (RFNBOs)	Greenhouse gas emission reduction from fuels	Limit for renewable liquid and gaseous fuels from food and feed crops	Limit for mature liquid and gaseous fuels from feedstocks IX.B	Multiple Counting
	(% cal)	(%)	(% cal)	(% cal)	(-)
2025	1.07	6 of which UER*) 1	7		x2 for advanced renewable fuels incl. RFNBO _s x2 for mature renewable fuels x4 for renewable electricity in road transport x1.5 for renewable electricity in rail transport x1.2 for renewable aviation and maritime fuels
2026	1.25	6.25	5.6	1.7	
2027	1.5	6.75			
2028	2.5	7.5			
2029	3.75	8.75			
2030 and onwards	5.5 RFNBO _s min 1	11			

Sources: FAS Prague, Association for Biodiesel Production based on the

- Act No. 42/2025 Coll., amending Act No. 201/2012 Coll., on air protection, as amended, and other related acts. The Act specifies the obligations of suppliers of liquid fuels.
- Act No. 87/2025 Coll., amending Act No. 458/2000 Coll., “Energy Act”, as amended, and other related acts, including Act No. 165/2012 Coll., on supported energy sources. The act specifies the obligations of gaseous fuels.

SANCTIONS:

- CZK 2 for each MJ of undelivered advanced liquid or gaseous renewable fuel, including fuels of non-biological origin RFNBO.
- CZK 10 for each kg of unreduced greenhouse gas emissions from fuels.

*) UER: Upstream emission reduction

Denmark

The following revision incorporates Denmark's shift from a volume-based blending mandate to a GHG emission reduction requirement, which was introduced in 2022 and updated to align with the European Union's revision of the second Renewable Energy Directive RED II.

Year	GHG Emission Reduction (%)	Advanced Biofuels (REDII Annex IX-A) (% caloric)	Crop-based Biofuel Cap	Multipliers
2025-2027	5.2 (3.4 in 2022-2024)	1.0	High-ILUC phased out (2025)	Advanced: 2x Road Electric: 4x Rail Electric: 1.5x Aviation and

				Maritime: 1.2x
2028-2029	6.0	1.0	TBD	Same as above
2030	7.0	3.5	TBD	Same as above

Source: [Danish Energy Agency](#)²

Key Policy Updates

Transition to GHG reduction targets: The target is a 5.2 percent GHG emission reduction for 2025-2027, rising to 7 percent by 2030. This mandate includes electricity used in transport.

Crop-based Biofuels: Biofuels from palm oil and its by-products (including palm fatty acid distillate (PFAD) and soybeans were phased out in 2022, unless certified as low- ILUC-risk. By 2025, all high-ILUC-risk biofuels must be phased out.

RED II Annex IX Feedstock Caps: The use of biofuels produced from Annex IX-B feedstocks are capped at 1.7 percent.

Multipliers: Denmark utilizes multipliers to incentivize specific technologies:

- 2x for advanced biofuels (Annex IX-A).
- 4x for renewable electricity in road transport.
- 1.5x for rail Electric.
- 1.2x for renewable fuels used in aviation and maritime sectors.

Penalties: Fuel suppliers that fail to meet their GHG reduction obligations are subject to fines and may face criminal liability under Danish law.

Estonia

	Overall Percentage (% cal)	Advanced Biofuels (% cal)	Cap on Advanced Biofuels from feedstocks listed in Part B of Annex IX (%)	Cap on crop-based biofuels (% cal)	Double Counting
2022	7.5	0.5	1.7	4.5	Yes
2023				2.5	
2024 - 2027				0.5	

Source: FAS Warsaw based on the Estonian *Liquid Fuel Act* passed on January 29, 2003, last amended in 2026

² <https://ens.dk/energikilder/transport/co2e-krav-til-braendstofleverandoerer>.

The blending obligation does not apply to 98-octane gasoline or fuel released to the market in emergency situations, mentioned under the Liquid Fuel Stocks Act or the Emergency Act.

Penalties

Failure to comply with these measures could result in the following penalties:

Committed by	Penalty
Natural person	Up to 300 fine units*
Legal entity	€125 for each gigajoule by which the legal entity falls short of fulfilling the obligation

*One fine unit currently is equal to €8. This rate is subject to change by the Estonian Penal Code.

Failure to comply with the obligation concerning the proportional blending can be fined with:

Committed by	Penalty
Natural person	Up to 300 fine units
Legal entity	Up to €100,000

Source: § 33¹ and § 33² of the Estonian *Liquid Fuel Act* passed on January 29, 2003, last amended in 2026.

Finland

The following revision reflects Finland's updated biofuels mandates as of March 2026. The mandates were moderated in late 2024 to manage fuel costs while still aligning with the revision of RED II targets for 2030.

Year	Overall Percentage (% cal)	Advanced (Annex IX-A) (% cal)	Crop-based Cap (% cal)	Multiple Counting
2025	16.5 (13.5 in 2024)	3.0	2.6	No
2026	19.5	4.0	2.6	No
2027	22.5	4.0	2.6	No
2028	31.0	8.0	2.6	No
2029	31.0	9.0	2.6	No
2030 and beyond	34.0	10.0	2.6	No

Sources: [Finnish Government](#)³, [Ministry of Economic Affairs and Employment of Finland](#)⁴, and [IEA Bioenergy](#)⁵.

³ <https://valtioneuvosto.fi/en/-/1410877/legislative-proposal-to-moderate-rise-in-distribution-obligation-levels-to-parliament>

⁴ <https://tem.fi/en/biofuels>

⁵ <https://www.iea.org/regions/europe>

Key Policy Updates

Mandate Moderation: In late 2024, Finland signed into law reduced distribution obligations for the 2025–2027 period to manage fuel price increases. The 2025 target was lowered from 29 percent to 16.5 percent, followed by 19.5 percent in 2026 and 22.5 percent in 2027.

Target Jump (2028-2030): To catch up with revised RED II goals, the mandate rises sharply to 31 percent in 2028 and reaches 34 percent by 2030.

Advanced Biofuels (Annex IX-A): Finland continues to prioritize advanced fuels, with a sub-target of 3 percent in 2025, rising to 10 percent by 2030.

Crop-based Restrictions: Biofuels from food and feed crops are capped at 2.6 percent. High-ILUC risk feedstocks (palm/soy) have been effectively phased out (0 percent) since 2021.

Annex IX-B Feedstocks: While waste-based feedstocks (e.g., used cooking oil) are not currently capped in Finland, the country is updating its laws to align with revised EU waste feedstock lists by late 2025.

Penalties:

Failure to comply with these measures could result in monetary penalties issued by the Energy Authority:

Mandate Type	Penalty Rate
General Biofuel Obligation	€0.04 per MJ (~€1,675/toe) of missing biofuel
Advanced Biofuel Sub-target	€0.03 per MJ (~€1,260/toe) of missing advanced biofuel

France

French decision makers prepared a new draft plan in 2025 that sets tougher goals for reducing GHG emissions in the transport sector in order to transpose the revised RED II directive into the national law but its full implementation is expected only in 2027.

In the draft,⁶ France directs fuel suppliers to cut the GHG intensity of transport fuels by:

- 7.1 percent in 2027
- 10.6 percent in 2030
- 18.7 percent in 2035

The draft also increases the minimum amount of biofuels that must be blended into diesel and gasoline, following revised RED II rules. In 2025–2026, France has not yet completed the transposition of revised RED II. During this interim period the French biofuel blending obligation system (TIRUERT in French) remains in force while the Carbon Intensity Reduction Requirement (IRICC in French) preparation is

⁶ Document in French [20250512 - projet_arrete_iricc.pdf](#)

underway, but IRICC⁷ obligations are not yet implemented. While TIRUERT applies only to road fuels, IRICC will also cover fuels used in the maritime, inland waterway, and aviation sectors, as well as compressed natural gas (CNG) and liquefied petroleum gas (LPG). France is expected to phase out the double-counting mechanism for Annex IX-A biofuels under the new regulatory framework. As a result, suppliers will no longer benefit from inflated compliance credits for limited volumes of advanced biofuels; instead, they will be required to meet their obligations based on actual renewable energy delivered. Under this system, the advanced-biofuel mandate for 2030 corresponds to a 1.95 percent reduction in greenhouse-gas intensity.

The latest updated version of the French Customs Code establishes the following mandates:

	Bioethanol (% cal)		Biodiesel (% cal)		Double Counting
Year	Total	Advanced (% cal)	Total	Advanced (% cal)	Yes
2026⁸	10.5	1.8	9.4	0.7	

Source: FAS Paris

Double counting for cellulosic biofuels and waste biofuels produced from the feedstocks listed in Annex IX of Directive 2009/28/EC except tall oil and tall oil pitch.

Cap on certain feedstocks

Since 2019 the share of energy that can be taken into account towards France's mandate is limited to a maximum of:

- 7 percent for conventional biofuels including biofuels produced from palm oil fatty acid distillates.
- 0.9 percent for used cooking oil and animal fats.
- 0.6 percent for tall oil and tall oil pitch.
- 0.2 percent for sugar plant residues and starch residues extracted from starch-rich plants (0.4 percent from 2020).
- Palm oil is excluded since January 1, 2020.
- Soybean oil is excluded since January 1, 2022.

Article 43 of the *Energy Transition Law for Green Growth* states that priority should be given to the development of advanced biofuels while preserving investments made in conventional biofuel production.

Penalties

Article 32 of the 2005 Finance Act introduced a tax (renamed *Taxe Incitative relative à l'Incorporation de Biocarburants* (Biofuel Incorporation Incentive Tax) in the 2019 Finance Act) penalizing operators who release a proportion of biofuels below blending targets. The operator is taxed on the difference

⁷ Document in French [20250512 - projet arrete iricc.pdf](#)

⁸Link in French [Article 266 quindecies - Code des douanes - Légifrance](#)

between the national target percentage of renewable energy incorporation and the proportion of renewable energy contained in their products.

In France, the General Directorate of Customs and Indirect Taxes (DGDDI) oversees fuel suppliers, while the General Directorate of Civil Aviation (DGAC) oversees aircraft operators and airport managers. Regulation (EU) 2023/2045, known as ReFuel EU Aviation⁹ also applies to aircraft operators who conduct at least 500 annual commercial passenger flights or 52 annual all-cargo flights from EU airports. These operators must refuel at EU airports to prevent competitive distortions and avoidance strategies.

The ReFuel EU Aviation regulation took effect on January 1, 2024, with reporting obligations for fuel suppliers and aircraft operators beginning in 2025. Reports cover SAF incorporations made in 2024, with deadlines on February 14 for fuel suppliers and March 31, 2025 for aircraft operators. The regulation includes a timeline for the minimum SAF incorporation in aviation fuel at EU airports, starting January 1, 2025.

Suppliers who do not meet these targets in France will face a fine at least equal to double the average price difference between SAF and traditional fuel, multiplied by the amount of fuel sold in excess of these thresholds. If the targets are not met, the supplier will have to not only meet the targets set for the following year but also compensate for previously accumulated deficits.¹⁰

Germany

Germany transposed the RED II directive into national law with the *Law for the Further Development of the Greenhouse Gas Reduction Mandate* of September 24, 2021. This law amends existing mandates and feedstock caps; introduces additional mandates for advanced biofuels and sustainable aviation fuel; and allows for additional compliance options. The national implementation of the revised RED II was officially published on June 5, 2026. Major changes include significant increases in the GHG emission reduction mandate, a stepwise increase of the cap on crop-based biofuel, and the end of double counting for advanced biofuels that exceed the subquota for Annex IX part A. UCO and Category 3 animal fat are not approved for co-processing HDRD. Additionally, only renewable energy sources used on road and rail may only be counted against the GHG reduction mandate.

Due to an amendment of the 38th Implementation Ordinance on the Federal Act on Protection against Air Pollution, the carryover of GHG reduction tickets from one year into the next is paused for 2025 and 2026. The surpluses may be carried over to 2027.

⁹ <https://www.ecologie.gouv.fr/politiques-publiques/carburants-daviation-durables-cad-sustainable-aviation-fuels-saf>.

¹⁰ [Carburants d'aviation durables \(CAD\) ou Sustainable aviation fuels \(SAF\) | Ministères Aménagement du territoire Transition écologique.](#)

	GHG Emission Reduction¹⁾ (%)	Advanced Biofuels^{2) 4)} (% cal)	RFNBO (% cal) ³⁾	Cap on crop-based biofuel²⁾ (% cal)	Cap on feedstocks with high ILUC risk^{2), 5)} (% cal)	Multiple counting
2026	12	2	0.1	4.9	0	See separate table
2027	17.5	3		4.9		
2028	19.5	3	0.5	4.6		
2029	22.5	3		4.7		
2030	26.5	3.5	1.5	4.9		
2031	30	4		5.0		
2032	33	4.5	3	5.5		
2033	36	5	3.5	5.8		
2034	38	5.5	4	-		
2035	41	6	5	-		
2036	46	6.5	6	-		
2037	51	7	7	-		
2038	56	7.5	8	-		
2039	61	8	9	-		
2040	65	9	10	-		

Sources: FAS Berlin based on [Federal Act on Protection against Air Pollution](#) , [38th Implementation Ordinance on the Federal Act on Protection against Air Pollution](#), [Upstream Emission Reduction Ordinance](#) and the [Second Law for the Amendment for the Further Development of the Greenhouse Gas Reduction Mandate](#) (all in German language)

- 1) *Federal Act on Protection against Air Pollution*
- 2) *38th Implementation Ordinance on the Federal Act on Protection against Air Pollution*, co-processing with UCO and cat. 3 animal fats as feedstock can no longer be counted towards the GHG reduction mandate.
- 3) *37th Implementation Ordinance on the Federal Act on Protection against Air Pollution*
- 4) Companies that put on the market 2 PJ or less of biofuels in the previous year are exempted.
- 5) Effectively, this means that since 2023, biofuels based on palm oil feedstock no longer count against the GHG reduction mandate, unless certified low ILUC-risk. Additionally, biofuels made from POME as a feedstock are no longer eligible for counting towards the GHG reduction mandate.

Multiple counting

Compliance Option	Conditions	Factor
Electricity ¹⁾	For road e-vehicles	3 ²

Sources: FAS Berlin based on

- 1) *38th Implementation Ordinance on the Federal Act on Protection against Air Pollution*, *Second Law for the Amendment for the Further Development of the Greenhouse Gas Reduction Mandate*
- 2) *Electricity will be triple counted through 2034, as of 2035 it will be double counted, and in 2036 it will no longer be double counted. Heavy electronic vehicles such as busses over 5 MT and trucks over 12 MT are eligible for quadruple counting through 2034, which will decrease stepwise as of 2035.*

Penalties

Failing to meet the mandates can result in the following penalties:

Mandate	Year	Penalty
GHG reduction	Since 2026	€0.60 per kg CO ₂ eq under allocated reduction
RFNBO	Since 2026	€120 per GJ under allocated
Advanced biofuels	Since 2026	€45 per GJ under allocated
SAF	Since 2026	€4,700 per MT under allocated
eSAF	2030	€17,000 per MT under allocated

Source: FAS Berlin based on *Federal Act on Protection against Air Pollution*

Greece

	Overall Percentage (% cal)	Biodiesel (% cal)	Bioethanol (% cal)	Advanced Biofuel	Multiple Counting
Since 2020	10.0	7.0	3.3	0.2	No

Source: FAS Rome

In 2012, *Decision 4062 (FEK 70/A/30.3.2012)* harmonized Greek legislation with European Commission Directive 2009/28/EC. As a result, the increased mandate of 10 percent in 2020 can be met either by domestic production or imports. The previous lower mandate was only allowed to be filled through domestic production.

[Law 3054/2002](#) and its amendments mandates that producers and distributors of petrol and diesel must blend their fuels with a quota of biofuels. The quota is specified in the “distribution scheme,” reviewed every year, and set at 7 percent for 2022 and 2023. This translates into 140 million liters for both 2022 and 2023. The distribution scheme for the last years has not been published.

Hungary

Year	Biodiesel (% cal)*	Bioethanol (% cal)**	Advanced Biofuels (% cal)	Cap on feedstocks with high ILUC risk (% cal)	Double Counting
2022 -2023	8.4	6.1	0.2	0.08	Biofuels derived from feedstock listed in Annex 2 of the Government Decree No. 821/2021 (in
2024	8.4	6.1	0.5	0.06	
2025	8.9	6.1	1	0.06	
2026	9.1	6.1	1	0.04	
2027	tbd	tbd	tbd	0.04	

2028-2029	tbd	tbd	tbd	0.02	Hungarian)
2030	tbd	tbd	5.5	0	

* In the case of diesel fuels, the share of hydrogenated vegetable oils must be at least 0.4 percent;

** The 6.1 percent share of renewables is applied to 95 octane gasoline of which maximum 1percent can be substituted with biomethanol or biomethyl-tert-butyl ether.

Source: FAS Budapest

Hungary has set a target minimum 25 percent share of renewable energy in transport by 2030. This obligation entered into force with the CXVII/2010 Act (in Hungarian) on promoting the use of renewable energy and the reduction of GHG emissions of energy used in transport. To reach this target, the overall share of advanced biofuels and renewable fuels of non-biological origin in final energy consumption will be increased to 5.5 percent in the sector by 2030, according to the updated [National Energy and Climate Plan](#). Additionally, a significant share of renewable energy targets is planned to be achieved through an increase in electric energy utilization as GHG emissions should be reduced in transportation to 0.9 million MT CO₂ equivalent by 2050.

Penalties

Failure to comply with these measures could result in the following penalty:

Mandate	Penalty
Energy % from biofuels	ca. € 88 per GJ under supplied

Ireland

	Overall Percentage (% vol)	Advanced biofuels & RFNBO (% cal)	Multiple Counting
2024	21	1.0	x2 for Annex IX biofuels; x2.5 for HVO/CHVO from Annex IX; x3.4 for BioCNG from Annex IX Part A; x4 for renewable electricity in road; x1.5 in rail; x1.2 for aviation and maritime fuels
2025	25	1.5	
2026	32	5.0 (combined)	
2027	36	11.0 (combined)	
2028	40	14.0 (combined)	
2029	44	18 (combined)	
2030	54	21.5 (advanced); 2.5 (RFNBO)	

Source: FAS London

Ireland's Renewable Transport Fuel Obligation (RTFO) is administered by the National Oil Reserves Agency (NORA) under Part 5A of the National Oil Reserves Agency Act 2007. The RTFO rate has increased significantly from 11 percent (by volume) in 2019 to 25 percent (by energy) in 2025, reflecting Ireland's commitment to achieving both Climate Action Plan targets and EU Renewable Energy Directive requirements.

Effective January 1, 2026, Ireland will extend the RTFO to include fuel supplied to rail transport. The country will also enable the award of RTFO certificates for renewable electricity supplied via public charge points to electric vehicles, with a multiplier of x4 to further incentivize electrification of the transport sector.

Bioethanol

Ireland began its E10 roll-out in April 2023 as part of measures to achieve the government target of a 51 percent reduction in transport emissions by 2030. The Climate Action Plan sets an ambition to maintain E10 blending through 2030.

In 2024, Ireland achieved a physical blend rate of 10.1 percent biofuel in road transport fuel, comprising a 9 percent ethanol blend in petrol and a 10 percent biodiesel/HVO blend in diesel. This performance puts Ireland on track to meet its Climate Action Plan 2025 interim targets of E10 and B12.

Advanced Biofuels & RFNBOs

Ireland has established a combined obligation for advanced biofuels (from Annex IX Part A feedstocks) and RFNBOs. The combined rate increases from 1.5 percent in 2025 to 5 percent in 2026, reaching a combined target where advanced biofuels must achieve 21.5 percent and RFNBOs 2.5 percent by 2030. This trajectory is designed to meet the revised RED II requirement of at least 5.5 percent combined advanced biofuels and RFNBOs by 2030, with a minimum 1 percent RFNBO share. A review of the combined obligation is planned for 2027 to determine the appropriate timing for introducing separate RFNBO obligations before 2030. This follows a downward revision to the current 2026 obligation from an initial target of 6.5 percent.

Ireland maintains a cap on crop-based biofuels at 7 percent (by energy content), though the country's actual consumption is significantly lower. Under revised RED II provisions, Ireland has adjusted its renewable energy share in transport (RES-T) target from 29 percent to 24 percent by 2030, reflecting its lower crop-based biofuel cap.

Penalties

Failing to meet the mandates can result in the following penalties:

Mandate	Penalty
Energetic	Overall Biofuel mandate: Buy-out price 0.05 €/MJ
	Advanced Biofuel mandate: Buy-out price 0.08 €/MJ

Source: ePure

Tax incentives

The Mineral Oil Tax (MOT) applies to fuels and is composed of a fuel excise and carbon tax (the MOTCC). The latter do not apply to biofuels.

Italy

Year	Overall Obligation (%)	Traditional Obligation (%)	Advanced Biofuels Obligation (%)	Bioethanol Obligation (%)	Advanced Biomethane Quota (%)	Advanced Biofuels Other Than Biomethane Quota (%)
2023	10	6.6	3.4	0.5	2.3	0.05
2024	10.8	6.6	4.2	1	2.9	0.05
2025	11.7	6.8	4.9	3	3.5	0.05
2026	12.6	7.1	5.5	3.4	3.9	0.05
2027	13.4	7.3	6.1	3.8	4.3	0.05
2028	14.3	7.6	6.7	4.2	4.8	0.05
2029	15.2	7.8	7.4	4.6	5.2	0.05
2030	16	8	8	5	5.7	0.00

Source: FAS Rome, based on a decree dated March 16, 2023, amended by a decree dated October 20, 2023, issued by Italy's Ministry of Environment and Energy Security

Penalties

As of 2020, fuel suppliers not complying with at least 95 percent of the renewable mandates have to pay a penalty of €750 per missing certificate. A certificate is equivalent to 10 Gcal $\approx$ 1 toe or 5 Gcal $\approx$ 0.5 toe for advanced biofuels. The penalty for advanced biofuels can be reduced if insufficient market availability is demonstrated.

Latvia

In accordance with the 2025 Transport Energy Law, which entered into force on January 1, 2026, GHG emissions generated by the transport sector will have to be reduced by at least 16 percent from 2030 onwards. Fuel suppliers will be required to increase the share of renewable energy in their energy portfolio. Each business operator will be able to choose the most effective measures for achieving set targets. For example, existing filling stations will have the option of installing biomethane or electricity charging points, as well as offer fuel with a higher biofuel blend or fully renewable fuel. In addition, the law sets new requirements for public transport in cities—at least 30 percent of public transport vehicles will have to use renewable energy. Existing vehicles may continue to be used if they are adapted to operate on cleaner energy, for example, by switching to biofuel, biomethane, or electricity.

In accordance with the transitional provisions of the 2025 Transport Energy Law, until December 31, 2029, 95-octane motor gasoline shall be sold in Latvia blended with minimal volume of 9.5 percent bioethanol or advanced biofuel in the total volume of the mixture. This obligation does not apply to 98-octane gasolines. Simultaneously, diesel shall be sold in Latvia blended with minimal volume of 6.5 percent biodiesel or advanced biofuel in the total volume of the mixture. Until March 31, 2026, it was permitted to sell diesel fuel with no biofuel added.

Tax incentives

The Latvian tax law supports the blending of biofuels by reducing the rate of excise duties. The following rates apply in 2025-2028.

Product	2025	2026	2028
Unleaded petrol per 1000 liters	532	555	555
Leaded petrol, per 1000 liters	617	640	640
Diesel, per 1000 liters	440.5	467*	467
Petrol with a high bioethanol content (from 70-85 percent by volume) (fuel E85), per 1000 liters	360	360	360
Diesel blended with biodiesel derived from biomass, per 1000 liters	440.5	467	467
Biodiesel (derived from biomass or paraffined diesel fuel derived from biomass) used as motor fuel, per 1000 liters	330	330	330
Biodiesel (derived from biomass or paraffined diesel fuel derived from biomass) used as heating fuel, per 1000 liters	21	21	21

Source: Latvian Ministry of Finance at <https://www.fm.gov.lv/lv/tax-rates-0>

*In connection with the geopolitical tensions in the Middle East and the resulting increase in the prices of petroleum products on international markets, in order to reduce the impact of the rise in fuel prices on Latvian entrepreneurs and residents, the Cabinet of Ministers to implement a number of measures, including supporting the temporary reduction of excise duty on diesel fuel from the previous €467 to €396 per 1000 liters, which entered into force from April 1, 2026 to June 30, 2026. The measure is intended to be a fiscally neutral stabilization mechanism for fuel prices for a limited period.

Lithuania

	Overall Percentage (% cal)	Advanced Biofuels (% cal)	Cap on crop-based biofuel (% cal)	Cap on UCO- and cat I and II animal fat-based biofuels (% cal)	Double counting
2022	6.8	0.2	No more	1.7	Yes

2023	7.2	0.4	than 1.0% higher than the total share in 2020		
2024	7.8	0.7			
2025	8.6	1.0			
2026	10	1.2			
2027	13	1.6			
2028	16	2.2			
2029	21	3.3			
2030	29	5.5			

Source: FAS Warsaw based on the Lithuanian *Law on Alternative Fuels* of 2021

Mandatory blending of biofuels into fossil fuels

Fuel suppliers in Lithuania must ensure that:

- Each litre of petrol contains a minimum of 6.6 percent by volume of bioethanol (a biofuel deficiency of 0.5 percent is permissible), calculated on the basis of the total energy value of the mixture of fuel and biofuel; blending into 98-octane petrol is optional,
- Each litre of diesel contains at least 6.2 percent by volume of biofuel (a biofuel deficiency of 0.5 percent is permissible), calculated on the basis of the total energy value of the mixture of fuel and biofuel.

On March 23, 2021, the Lithuanian parliament approved the *Law on Alternative Fuels* (LAF). Under the law, the transport sector is encouraged to shift to electricity, biomethane, and hydrogen, increasing the requirements for blending biofuels. The LAF established clear directions for the development of alternative fuel vehicles and the infrastructure required for them.

The LAF introduced progressively increasing obligations for fuel suppliers regarding the use of biofuels, which will be possible to implement more flexibly. In order to encourage the use of biomethane and other advanced biofuels and hydrogen, their blending is offset by twice the energy value.

The Netherlands

The following revision reflects the Netherlands' finalized transition from an energy-based to a GHG reduction-based system, effective January 1, 2026, as mandated by the transposition of the revised RED II.

Year	Overall Target	Advanced (REDII Annex IX-A)	Crop-based Cap	Annex REDII IX-B Cap	Unit Base
2023	18.9	2.4	1.4	10 ^a	% caloric
2024	28.4	2.9	1.4	10 ^a	% caloric
2025	29.4	3.6	1.4	10 ^a	% caloric

2026	14.4	3.1	1.2	4.3	% GHG red.
2027	16.4	4.5	1.2	4.3	% GHG red.
2028	22.8	5.9	1.2	4.3	% GHG red.
2029	24.8	7.3	1.2	4.3	% GHG red.
2030	27.1	8.8	1.2	4.3	% GHG red.

(a) With double counting. Sources: [Dutch Emissions Authority \(NEA\)](#)¹¹ and [Dutch Ministry of Infrastructure and Water Management](#)¹².

Key Policy Updates

Transition to GHG reduction targets: Starting January 1, 2026, the Netherlands will shift from an energy-based mandates (% caloric) to a GHG reduction-based system (% GHG reduction).

Target Adjustments: The Dutch government retroactively increased the 2024 target to 28.4 percent cal (from 19.9 percent) and the 2025 target to 29.4 percent cal. From 2026, the target is 14.4 percent GHG reduction, rising linearly to 27.1 percent by 2030.

Sector Expansion: The new Fuel Transition Obligation ([Brandstoftransitieverplichting](#)) now includes mandatory targets for inland and maritime shipping, which were previously voluntary opt-in sectors.

Feedstock Restrictions

Conventional Biofuels: Capped at 1.2 percent for land transport from 2026 onwards.

Palm and Soy based biofuels: Banned except for certified low-Indirect Land Use Change (ILUC-risk) feedstock.

Revised RED II Annex IX-B: Limited to 4.3 percent in the land transport sector.

Penalties: Fuel suppliers failing to meet their annual obligation are subject to fines issued by the Dutch Emissions Authority (NEA), which can result in court proceedings for continued non-compliance.

Poland

The amendment of the *Law on Biocomponents and Liquid Biofuels* was published on March 12, 2025. In the new provisions the National Indicative Target (NIT) for use of renewable energy sources (RES) in transport was established for 2026 and beyond: NIT will increase to 10 percent in 2026-2029, and to 14.9 percent in 2030 and subsequent years. The higher shares of RES in transport will be achieved primarily by using biocomponents derived from food or feed raw materials, advanced biocomponents produced from waste raw materials or residues, and renewable electricity used in the transport sector. Additionally, solid and gas fuels from biomass used in energy sectors outside transport will also be able to count for NIT. The new regulations will allow for the development of alternative fuels, such as

¹¹ <https://www.emissieautoriteit.nl/regelgeving/hernieuwbare-energie-voor-vervoer-hbe/verplichtingen>

¹² <https://www.government.nl/ministries/ministry-of-infrastructure-and-water-management>

biomethane, as well as, among others, electricity from renewable energy sources used in the transport sector. Moreover, capping of agricultural inputs at 6.1 percent for production of biofuels was introduced.

The general amendment entered into force on April 1, 2025; implementation of the NIT will enter into force on January 1, 2026.

	Overall Percentage (% cal)	Biodiesel (% cal)	Bioethanol (% cal)	Cap on conventional crop-based biofuel (% cal)
2022	8.8	5.2	3.2	7
2023	8.9	5.2	3.2	7
2024	9.1	5.2	3.2	7
2025	9.2	5.2	4.59	7
2026	10	6.2	4.59	6.1
2027	10	6.2	4.59	6.1
2028	10	6.2	4.59	6.1
2029	10	6.2	4.59	6.1
2030	14.9	6.2	4.59	6.1

Source: FAS Warsaw based on the Polish *Act on Bio-components and Liquid Biofuels* as amended in March 2025.

For advanced biofuels, Polish legislation was aligned with RED II feedstock listed in Part A and Part B of Annex IX. This included minimal use of advanced biofuels produced from feedstock listed in Part A of Annex IX and a cap on use of advanced biofuels produced from feedstock listed in Part B of Annex IX.

Advanced Biofuel Sources, Part A and Part B of Annex IX, Mandates and Cap		
	Part A Mandates (% cal)	Part B Cap (% cal)
2026	1	3.4
2027	1	3.4
2028	1	3.4
2029	1	3.4
2030	3.5	3.4

Source: FAS Warsaw based on the Polish *Act on Bio-components and Liquid Biofuels* as amended in March 2025.

Additionally, effective January 1, 2026, multiple counting for implementation of the NIT was introduced, as below:

- 1) electricity from renewable energy sources shall be deemed to be equal to 4 times its energy value – if supplied by charging;

- 2) electricity from renewable energy sources shall be deemed to be equal to 1.5 times its energy value – if supplied to rail vehicles;
- 3) advanced biofuels produced from feedstock specified in Annex IX of RED II shall be deemed to be equal to 2 times their energy value;
- 4) biofuels, other renewable fuels, liquid carbon fuels from recycling and gaseous carbon fuels from recycling contained in aviation fuels and marine fuels shall be deemed to be equal to 1.2 times their energy value.

Effective January 1, 2024, an amendment to the *Law on Biocomponents and Liquid Biofuels* introduced a requirement that 95-octane motor gasoline shall be sold in Poland in the format of E10, while 98-octane gasolines will remain in the E5 format. For this, mandatory blending for 95-octane motor gasoline at 5.3 percent (and detailing that 4.59 percent should be realized with bioethanol) was set, while for 98-octane gasoline, the level of mandatory blending remained unchanged at 3.2 percent.

Portugal

Portugal has introduced multiple counting. In addition to the country's biofuel consumption mandate and double counting for biofuels produced out of Annex IX feedstock, the country has put in place an x2 for fuels of non-biological origin, an x5 multiplier for electricity in road transport, and an x2.5 percent for renewable energy in train transportation.

	Overall Percentage (% cal)	Annex IX (% cal)	Cap on conventional crop-based biofuel (% cal)	Double Counting
2022	11	0.2	3.1	Yes
2023	11.5	0.7		
2024	11.5	0.7		
2025-2026	13	2		
2027-2028	14	4		
2029	16	7		
2030	16	10		

Sources: FAS Madrid based on consumption mandates: [Decree-Law 117/2010](#), [Decree-Law 69/2016](#), [Law 42/2016](#), *Budget Law* for 2018 and 2019 and [Decree-Law 8/2021](#) as amended by [Rectification Declaration 9-A/2021](#), [Decree-Law 84/2022](#), and [Decree-Law 23/2023](#). Double counting: [Decree-Law 117/2010](#) and Annex III in [Implementing Order 8/2012](#).

Penalties

Failure to comply with these measures could result in the following penalties:

Year	Penalty
2011 and onwards	€2,000 per TdB (Biofuel Entitlement, equals a Ktoe) that the obliged party fails to meet.

Source: FAS Madrid based on [Implementing Regulation 301/2011](#).

For additional information about Portugal’s biofuel sector, see the GAIN Report: *Portugal Biofuels Policy and Market* available through the FAS report database at <https://gain.fas.usda.gov/#/search>.

Romania

	Overall Percentage (% vol)	Biodiesel (% vol)	Bioethanol (% vol)	Double counting
2021 and onwards	10	6.5	8.0 (at least 2 percent April – June 2026)	Yes

Sources: FAS Bucharest based on *Government Decisions 1121/2013* and *931/2017*

Romania transposed RED II into Romanian national legislation through the *Emergency Ordinance (E.O.) 163/2022*. The EU directive 2023/2413 was transposed in 2025 through Emergency Ordinance 59/2025. Though it is fully implemented, the Emergency Ordinance must be subject to debate within the Romanian Parliament to become law. The draft passed the Senate approval, and it is under debate in the Chamber of Deputies, hence some amendments might affect the original piece of law. In April 2026, in the context of rising fuel prices and instituting the “crisis situation” state for 3 months (April – June 2026), the Romanian government changed the minimum blending requirement for bioethanol from 8 percent to 2 percent, leaving the biodiesel mandate unchanged.

Penalties

Failure to comply with these measures could result in the following penalties:

Year	Penalty
2019 and onwards	RON 70,000-100,000 (approx. USD 15,000-22,000)

Source: FAS Bucharest based on provisions of *Emergency Ordinance 80/2018*

Slovakia

Slovakia amended Act No. 309/2009 Coll. in 2025 to transpose revised RED II. The revised framework keeps the existing minimum blending obligations for gasoline and diesel, while increasing the overall renewable energy target in transport, especially after 2027. Compliance may increasingly rely on credit-based mechanisms, renewable electricity, biomethane, and other renewable fuels.

	Overall Percentage (% cal)	Bioethanol (% vol)	Biodiesel (% vol)	Advanced Biofuels (% cal)	Double Counting
2021	8.0	Minimum E9¹⁾	Minimum B6.9¹⁾	0.3	Yes
2022	8.2				

2023	8.6			0.5	(expanded under revised RED II framework from 2026 to include renewable electricity, biomethane, RFNBOs, and pure renewable fuels)
2024	8.8			0.65	
2025	9.2			1.05	
2026	9.5				
2027	10.2			1.25	
2028	11.5			1.75	
2029	16			2.75 + RFNBO min. 0,5%	
2030	27				

Source: FAS Prague based on Act No. 309/2009 Coll. on the Support of Renewable Energy Sources and High Efficiency Cogeneration, as amended by Act No. 259/2025 Coll., and related legislation.

Note: Minimum blending obligations remain at E9 for gasoline and B6.9 for diesel. The revised framework expands compliance options, including renewable electricity, biomethane, RFNBOs, and pure renewable fuels.

Slovenia

	Overall Percentage (% cal)	Biodiesel/ Bioethanol	Advanced Biofuels (% cal)	Cap on crop-based biofuels (% cal)
2023	10.3	Not specified	0.2	7
2024	10.6		0.2	
2025-2026	10		1	
2030	26		5.5	

Source: Act on the Promotion of the Use of Renewable Energy Sources ([ZSROVE-1](#))

The Act on the Promotion of the Use of Renewable Energy Sources (ZSROVE-1) was published on December 12, 2025, and entered into force on December 29, 2025.

This new regulation transposes the modified Directive (EU) 2018/2001 into national law. It mandates that the share of energy from renewable sources in final energy consumption for transport must reach at least 26 percent by 2030. Meanwhile, the minimum annual share may vary but cannot fall below 10 percent.

The share of advanced biofuels and biogas produced from feedstocks listed in Part A of Annex IX to Directive 2018/2001 must be at least 1 percent and rise to 5.5 percent by 2030 in final energy consumption for transport. The total share of crop-based raw materials may not exceed 7 percent.

The share of biofuels produced from raw materials listed in Part B of Annex IX to Directive 2018/2001 - in the energy content of fuels supplied to the transport sector - must not exceed 1.7 percent by 2030.

Penalties

A fine of EUR 5,000 to EUR 125,000 shall be imposed on a legal person if it fails to achieve the target share of renewables. The fine imposed on a legal entity shall be EUR 15,000 to EUR 250,000 if it qualifies as a medium-sized or large company.

A fine of EUR 2,000 to EUR 10,000 shall be imposed on a sole proprietor or an individual who independently carries out an activity in the case non-compliance. The fine shall be EUR 1,000 to EUR 5,000 if the non-compliant person is only a responsible representative of a legal entity or a sole proprietor.

Spain

Spain has established a 28 percent target of renewable energy in transport for 2030, with an intermediate goal in 2025 for 15 percent. In addition to biofuel consumption mandates and double counting for those biofuels produced out of Annex IX feedstock, the country has put in place an x4 multiplier for electricity in road transport, an x1.5 multiplier for renewable energy in trains and, as of 2024, an x1.2 multiplier for renewable marine and aviation transport, provided the feedstocks are of non-crop origin. Similarly, from 2025 onward, fuels of non-biological origin may be counted towards the mandate and be subject to double counting.

	Overall Percentage (% cal)	Annex IX- Part A (% cal)	Annex IX- Part B (% cal)	High ILUC Risk Biofuels (% cal)	Cap on crop-based biofuels (% cal)	Double counting
2022	10	0.2	1.7	3.1	7	Yes
2023	10.5	0.3	1.7	3.1	3.5	
2024	11	0.5	1.7	1.5	3.1	
2025	11.5	1	1.7	0	2.6	
2026	14	1.2	1.7	0	2.6	
2030	14	5.5	1.7	0	2.6	

Source: FAS Madrid based on [Royal Decree-Law 4/2013](#), [Royal Decree 1085/2015](#), [Royal Decree 205/2021](#), [Resolution of 30 of March 2021](#), [Resolution of 29 of September 2021](#), [Royal Decree 376/2022](#), [Ministerial Order TED 1342/2022](#), Ministerial [Order TED 728/2024](#), and [Royal Decree 5/2026](#).

Penalties

Failure to comply with these measures could result in the following penalties:

Year	Penalty
Since 2022	€1,623 per missing certificate (each certificate equals one Ktoe.)

Source: FAS Madrid based on [Resolution of 17 of December of 2021](#) and Ministerial Order [TED 728/2024](#) by the Ministry for Ecological Transition and Demographic Challenge.by the Ministry for Ecological Transition and Demographic Challenge.

[Royal Legislative Decree 6/2022](#) established the obligation to reduce GHG emissions by fossil fuels by 6 percent as of 2022 compared to 94.1 g CO₂ equivalent/MJ in 2010. The GHG emissions calculation methodology is laid out by [Royal Decree 235/2018](#).

Additional details regarding mandates are available in the draft [Royal Decree](#) on renewable fuels promotion (Spanish language only), submitted to public comment between January 9, 2026 and January 19, 2026.

For additional information about Spain’s biofuels sector, see GAIN Report *Spain Biofuels Policy and Market* available through the FAS report database at <https://gain.fas.usda.gov/#/search> .

Sweden

The following revision incorporates Sweden’s recent policy shift to increase mandate levels starting in mid-2025, following a period of significant reductions. It also aligns with the latest implementation strategies for the revised RED II, moving away from previous plans to abolish the quota entirely.

Year	Gasoline GHG Reduction	Diesel GHG Reduction	Unit Base
2023	7.8	30.5	% GHG reduction
2024 – June 2025	6.0	6.0	% GHG reduction
July 2025 – 2030	10.0	10.0	% GHG reduction

Sources: [Swedish Energy Agency](#)¹³, [Research Institute of Sweden](#)¹⁴, and [Svebio](#)¹⁵.

Key Policy Updates

Mandate Reversal (2025): After reducing the mandates to a 6 percent in 2024, the Swedish government announced in late 2024 that it will increase the obligation to 10 percent for both gasoline and diesel starting July 1, 2025. This update replaces the earlier proposal to abolish the quota in 2027.

Alignment with the revised RED II: To meet EU-wide targets, the Swedish Energy Agency has indicated that higher electrification alone is insufficient. Consequently, the mandate will likely remain at or above 10 percent through 2030 to ensure Sweden reaches its 70 percent transport emission reduction goal compared to 2010 levels.

Crop-based Biofuels: While there are no explicit national volume targets to limit crop-based fuels, High-Indirect Land Use Change (ILUC)-risk feedstocks remain excluded from the reduction quota unless certified as low-ILUC-risk.

High-Blend Incentives: Fuels like HVO100, E85, and FAME100 continue to operate outside the primary reduction mandate. These are tax reductions or exemptions managed by the [Swedish Tax Agency](#) to encourage uptake beyond the standard blend wall.

Penalties: Fuel suppliers that fail to meet their GHG reduction obligations are liable for a penalty fee of up to SEK 7 per kg CO₂-equivalent of shortfall.

Related reports: Please check the FAS report database for related reports at <https://gain.fas.usda.gov/#/search> .

¹³ <https://www.energimyndigheten.se/en/climate/sustainable-fuels/greenhouse-gas-reduction-mandate/>

¹⁴ <https://www.ri.se/en/biorefinery/fossil-free-fuels/blogpost/what-you-need-to-know-about-the-swedish-biofuel-reduction>

¹⁵ <https://www.svebio.se/en/>

Attachments:

No Attachments.